Phonology

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Not to be confused with Phenology.
For the study of language production and perception, see Phonetics. For other uses, see Phonology (disambiguation).

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Phonology (formerly also **phonemics** or **phonematics**[1][2][3][4][a]) is the branch of linguistics that studies how languages systematically organize their phonemes or, for sign languages, their constituent parts of signs. The term can also refer specifically to the sound or sign system of a particular language variety. At one time, the study of phonology related only to the study of the systems of phonemes in spoken languages, but now it may relate to any linguistic analysis either:

a. at a level beneath the word (including syllable, onset and rime, articulatory gestures, articulatory features, mora, etc.), or
b. all levels of language in which sound or signs are structured to convey linguistic meaning.[9]

Sign languages have a phonological system equivalent to the system of sounds in spoken languages. The building blocks of signs are specifications for movement, location, and handshape.[10] At first, a separate terminology was used for the study of sign phonology ("chereme" instead of "phoneme", etc.), but the concepts are now considered to apply universally to all human languages.

Terminology [edit]

The word "phonology" (as in "phonology of English") can refer either to the field of study or to the phonological system of a given language.[11] This is one of the fundamental systems that a language is considered to comprise, like its syntax, its morphology and its lexicon. The word *phonology* comes from Ancient Greek φωνή, *phōnḗ*, 'voice, sound', and the suffix *-logy* (which is from Greek λόγος, *lógos*, 'word, speech, subject of discussion').

Phonology is typically distinguished from phonetics, which concerns the physical production, acoustic transmission and perception of the sounds or signs of language.[12][13] Phonology describes the way they function within a given language or across languages to encode meaning. For many linguists, phonetics belongs to descriptive linguistics and phonology to theoretical linguistics, but establishing the phonological system of a language is necessarily an application of theoretical principles to analysis of phonetic evidence in some theories. The distinction was not always made, particularly before the development of the modern concept of the phoneme in the mid-20th century. Some subfields of modern phonology have a crossover with phonetics in descriptive disciplines such as psycholinguistics and speech perception, which result in specific areas like articulatory phonology or laboratory phonology.

Definitions of the field of phonology vary. Nikolai Trubetzkoy in *Grundzüge der Phonologie* (1939) defines phonology as "the study of sound pertaining to the system of language," as opposed to phonetics, which is "the study of sound pertaining to the act of speech" (the distinction between *language* and *speech* being basically Ferdinand de Saussure's distinction between *langue and parole*).[14] More recently, Lass (1998) writes that phonology refers broadly to the subdiscipline of linguistics concerned with the sounds of language, and in more narrow terms, "phonology proper is concerned with the function, behavior and organization of sounds as linguistic items."[12] According to Clark *et al.* (2007), it means the systematic use of sound to encode meaning in any spoken human language, or the field of linguistics studying that use.[15]

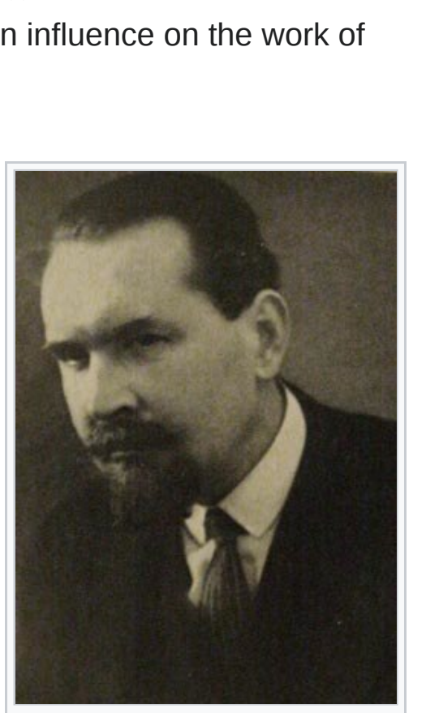
History [edit]

Evidence for a systematic investigation of the sounds of a language appears in the 4th c. BCE Ashtadhyayi, a Sanskrit grammar by Pāṇini. Particularly, within the Shiva Sutras, auxiliary work to the Ashtadhyayi, an inventory of what would be construed as a list of the phonemes of Sanskrit is provided, with a notational scheme for them which is deployed throughout the main text, which concern itself with issues of morphology, syntax and semantics.

Ibn Jinni of Mosul, a pioneer in phonology, wrote prolifically in the 10th century on Arabic morphology and phonology in works such as *Kitāb Al-Munṣif*, *Kitāb Al-Muḥtasab*, and Kitāb Al-Khaṣāʾiṣ [ar].[16]

The study of phonology as it exists today is defined by the formative studies of the 19th-century Polish scholar Jan Baudouin de Courtenay,[17]:17 who (together with his students Mikołaj Kruszewski and Lev Shcherba in the Kazan School) shaped the modern usage of the term *phoneme* in a series of lectures in 1876–1877. The word *phoneme* had been coined a few years earlier, in 1873, by the French linguist A. Dufriche-Desgenettes. In a paper read at 24 May meeting of the Société de Linguistique de Paris,[18] Dufriche-Desgenettes proposed for *phoneme* to serve as a one-word equivalent for the German *Sprachlaut*.[19] Baudouin de Courtenay's subsequent work, though often unacknowledged, is considered to be the starting point of modern phonology. He also worked on the theory of phonetic alternations (what is now called allophony and morphophonology) and may have had an influence on the work of Saussure, according to E. F. K. Koerner.[20]

An influential school of phonology in the interwar period was the Prague school. One of its leading members was Prince Nikolai Trubetzkoy, whose *Grundzüge der Phonologie* (*Principles of Phonology*),[14] published posthumously in 1939, is among the most important works in the field from that period. Directly influenced by Baudouin de Courtenay, Trubetzkoy is considered the founder of morphophonology, but the concept had also been recognized by de Courtenay. Trubetzkoy also developed the concept of the *archiphoneme*. Another important figure in the Prague school was Roman Jakobson, one of the most prominent linguists of the 20th century. Louis Hjelmslev's glossematics also contributed with a focus on linguistic structure independent of phonetic realization or semantics.[17]:175


Nikolai Trubetzkoy, 1920s

In 1968, Noam Chomsky and Morris Halle published *The Sound Pattern of English* (SPE), the basis for generative phonology. In that view, phonological representations are sequences of segments made up of distinctive features. The features were an expansion of earlier work by Roman Jakobson, Gunnar Fant, and Morris Halle. The features describe aspects of articulation and perception, are from a universally fixed set and have the binary values + or −. There are at least two levels of representation: underlying representation and surface phonetic representation. Ordered phonological rules govern how underlying representation is transformed into the actual pronunciation (the so-called surface form). An important consequence of the influence SPE had on phonological theory was the downplaying of the syllable and the emphasis on segments. Furthermore, the generativists folded morphophonology into phonology, which both solved and created problems.

Natural phonology is a theory based on the publications of its proponent David Stampe in 1969 and, more explicitly, in 1979. In this view, phonology is based on a set of universal phonological processes that interact with one another; those that are active and those that are suppressed is language-specific. Rather than acting on segments, phonological processes act on distinctive features within prosodic groups. Prosodic groups can be as small as a part of a syllable or as large as an entire utterance. Phonological processes are unordered with respect to each other and apply simultaneously, but the output of one process may be the input to another. The second most prominent natural phonologist is Patricia Donegan, Stampe's wife; there are many natural phonologists in Europe and a few in the US, such as Geoffrey Nathan. The principles of natural phonology were extended to morphology by Wolfgang U. Dressler, who founded natural morphology.

In 1976, John Goldsmith introduced autosegmental phonology. Phonological phenomena are no longer seen as operating on *one* linear sequence of segments, called phonemes or feature combinations but rather as involving *some parallel sequences* of features that reside on multiple tiers. Autosegmental phonology later evolved into feature geometry, which became the standard theory of representation for theories of the organization of phonology as different as lexical phonology and optimality theory.

Government phonology, which originated in the early 1980s as an attempt to unify theoretical notions of syntactic and phonological structures, is based on the notion that all languages necessarily follow a small set of principles and vary according to their selection of certain binary parameters. That is, all languages' phonological structures are essentially the same, but there is restricted variation that accounts for differences in surface realizations. Principles are held to be inviolable, but parameters may sometimes come into conflict. Prominent figures in this field include Jonathan Kaye, Jean Lowenstamm, Jean-Roger Vergnaud, Monik Charette, and John Harris.

In a course at the LSA summer institute in 1991, Alan Prince and Paul Smolensky developed optimality theory, an overall architecture for phonology according to which languages choose a pronunciation of a word that best satisfies a list of constraints ordered by importance; a lower-ranked constraint can be violated when the violation is necessary in order to obey a higher-ranked constraint. The approach was soon extended to morphology by John McCarthy and Alan Prince and has become a dominant trend in phonology. The appeal to phonetic grounding of constraints and representational elements (e.g. features) in various approaches has been criticized by proponents of "substance-free phonology", especially by Mark Hale and Charles Reiss.[21][22]

An integrated approach to phonological theory that combines synchronic and diachronic accounts to sound patterns was initiated with Evolutionary Phonology in recent years.[23]

Analysis of phonemes [edit]

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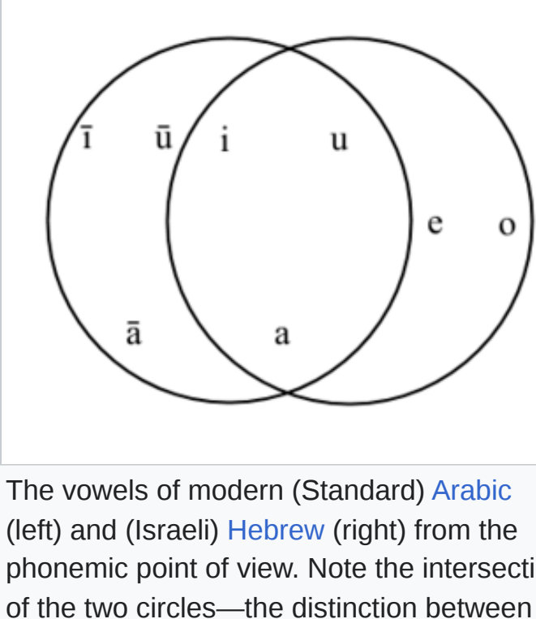
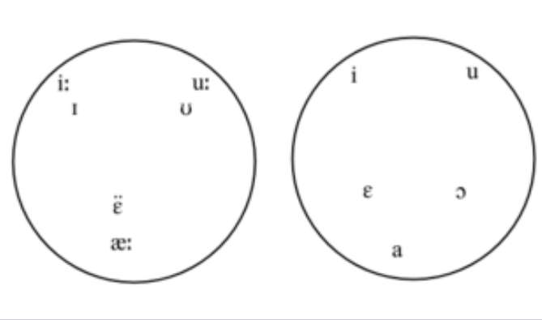
This section contains **phonetic transcriptions** in the **International Phonetic Alphabet (IPA)**. For an introductory guide on IPA symbols, see Help:IPA. For the distinction between [ ], / / and ⟨ ⟩, see IPA § Brackets and transcription delimiters.

An important part of traditional, pre-generative schools of phonology is studying which sounds can be grouped into distinctive units within a language; these units are known as phonemes. For example, in English, the "p" sound in *pot* is aspirated (pronounced [pʰ]) while that in *spot* is not aspirated (pronounced [p]). However, English speakers intuitively treat both sounds as variations (allophones, which cannot give origin to minimal pairs) of the same phonological category, that is of the phoneme /p/. (Traditionally, it would be argued that if an aspirated [pʰ] were interchanged with the unaspirated [p] in *spot*, native speakers of English would still hear the same words; that is, the two sounds are perceived as "the same" /p/.) In some other languages, however, these two sounds are perceived as different, and they are consequently assigned to different phonemes. For example, in Thai, Bengali, and Quechua, there are minimal pairs of words for which aspiration is the only contrasting feature (two words can have different meanings but with the only difference in pronunciation being that one has an aspirated sound where the other has an unaspirated one).



The vowels of modern (Standard) Arabic (left) and (Israeli) Hebrew (right) from the phonemic point of view. Note the intersection of the two circles—the distinction between short *a*, *i* and *u* is made by both speakers, but Arabic lacks the mid articulation of short vowels, while Hebrew lacks the distinction of vowel length.



The vowels of modern (Standard) Arabic and (Israeli) Hebrew from the phonetic point of view. The two circles are totally separate—none of the vowel-sounds made by speakers of one language is made by speakers of the other.

Part of the phonological study of a language therefore involves looking at data (phonetic transcriptions of the speech of native speakers) and trying to deduce what the underlying phonemes are and what the sound inventory of the language is. The presence or absence of minimal pairs, as mentioned above, is a frequently used criterion for deciding whether two sounds should be assigned to the same phoneme. However, other considerations often need to be taken into account as well.

The particular contrasts which are phonemic in a language can change over time. At one time, [f] and [v], two sounds that have the same place and manner of articulation and differ in voicing only, were allophones of the same phoneme in English, but later came to belong to separate phonemes. This is one of the main factors of historical change of languages as described in historical linguistics.

The findings and insights of speech perception and articulation research complicate the traditional and somewhat intuitive idea of interchangeable allophones being perceived as the same phoneme. First, interchanged allophones of the same phoneme can result in unrecognizable words. Second, actual speech, even at a word level, is highly co-articulated, so it is problematic to expect to be able to splice words into simple segments without affecting speech perception.

Different linguists therefore take different approaches to the problem of assigning sounds to phonemes. For example, they differ in the extent to which they require allophones to be phonetically similar. There are also differing ideas as to whether this grouping of sounds is purely a tool for linguistic analysis, or reflects an actual process in the way the human brain processes a language.

Since the early 1960s, theoretical linguists have moved away from the traditional concept of a phoneme, preferring to consider basic units at a more abstract level, as a component of morphemes; these units can be called *morphophonemes*, and analysis using this approach is called morphophonology.

Other topics [edit]

In addition to the minimal units that can serve the purpose of differentiating meaning (the phonemes), phonology studies how sounds alternate, or replace one another in different forms of the same morpheme (allomorphs), as well as, for example, syllable structure, stress, feature geometry, tone, and intonation.

Phonology also includes topics such as phonotactics (the phonological constraints on what sounds can appear in what positions in a given language) and phonological alternation (how the pronunciation of a sound changes through the application of phonological rules, sometimes in a given order that can be feeding or bleeding,[24]) as well as prosody, the study of suprasegmentals and topics such as stress and intonation.

The principles of phonological analysis can be applied independently of modality because they are designed to serve as general analytical tools, not language-specific ones. The same principles have been applied to the analysis of sign languages (see Phonemes in sign languages), even though the sublexical units are not instantiated as speech sounds.

See also [edit]

- Absolute neutralisation
- Accent (sociolinguistics)
- Cherology
- English phonology
- List of phonologists
- Neogrammarian
- Phonological development
- Phonological hierarchy
- Second language phonology

Notes [edit]

a. ^ Depending on usage, there may or may not be a distinction between *phonology* and *phonemics*/*phonematics*, the latter of which are more strictly about segments (consonants and vowels) rather than suprasegmentals (syllables, length, accent, tone, intonation, etc.). Synonymy between these terms is favored by some American structuralist phonologists (or rather, phonemicists)[2][3][5][6][7][8] and is reflected in *morphophonology*/morphophonemics, among other derivatives.

References [edit]

1. ^ Buck, Carl Darling (January 1945). ""Phonemics" versus "Phonematics"". *Classical Philology*. **40** (1). University of Chicago: 47. "Professor Whatmough's assault on the terms "phonemic" and "phonemics" seemed to me, as to others, pedantic. So it is a pleasure to discover that his antipathy to these now well-established terms is not so deep rooted and consistent as one would suppose from his words in *CP*, XXXVIII (1943), 211: "Nobody says *mathemics* instead of *mathematics*; and I, for one, do not say, and never shall, *phonemics* for *phonematics* or *phonemic* for *phonematic*." In happening to re-read an earlier article of his in the *Mélanges linguistiques offerts à M. Holger Pedersen* (1937),. I find him beginning a sentence (p. 46) "Ideally the phonemic system of a language. . . . ." "
2. ^ *a b* Trask (1996, p. 267): "**phonemics** /fəˈniːmɪks/ *n.* [*obsolescent*] 1. Any procedure for identifying the **phonemes** of a language from a corpus of data. 2. (formerly also **phonematics**) A former synonym for **phonology**, often preferred by the **American Structuralists** and reflecting the importance in structuralist work of phonemics in sense 1."
3. ^ *a b* Trask (1996, p. 264): "**phonematics** /fəʊnɪˈmætɪks/ *n.* 1. [*obsolete*] An old synonym for **phonemics** (sense 2)."
4. ^ "phonematics". *Collins*.
5. ^ Trauth, Gregory; Kazzazi, Kerstin, eds. (1998). "phonemics". *Hadumod Bussmann Lexikon der Sprachwissenschaft* [*Routledge Dictionary of Language and Linguistics*]. Translated by Trauth, Gregory; Kazzazi, Kerstin. Routledge. p. 892. "1 Synonym for **phonology**.
2 Because of the historical connotations that since the time of the Neogrammarians were attached to the term **phonology**, which today is used for synchronic and diachronic studies, 'phonemics' was first used by the American structuralists for 'synchronic phonology.' This designation was also meant to distinguish the American structuralist approach from that of the European structuralists, especially those of the **Prague School**."
6. ^ Crystal, David, ed. (2008). "phoneme". *A Dictionary of Linguistics and Phonetics* (6th ed.). Blackwell Publishing. p. 362. ISBN 978-1-4051-5296-9. "On this general basis, several approaches to **phonemic analysis**, or **phonemics**, have developed. [...] Apart from the question of definition, if the view is taken that all aspects of the sound system of a language can be analysed in terms of phonemes – that is, the suprasegmental as well as the segmental features – then 'phonemics' becomes equivalent to phonology (= **phonemic phonology**). This view was particularly common in later developments of the American structuralist tradition of linguistic analysis, where linguists adopting this 'phonemic principle' were called **phonemicists**. Many phonologists, however (particularly in the British tradition), prefer not to analyse suprasegmental features in terms of phonemes, and have developed approaches which do without the phoneme altogether ('nonphonemic phonology', as in prosodic and distinctive feature theories)."
7. ^ Richards, Jack Croft; Schmidt, Richard, eds. (2010). "**phonemics** *n* **phonemic** *adj*". *Longman Dictionary of Language Teaching & Applied Linguistics* (4th ed.). Pearson Education. p. 433. ISBN 978-1-4082-0460-3. "The term "phonemics" has been used by American linguists, particularly in structural linguistics. Lately, the term phonology has been preferred."
8. ^ Brown, Keith; Miller, Jim, eds. (2013). "phonemics". *The Cambridge Dictionary of Linguistics*. Cambridge University Press. p. 340. ISBN 978-0-521-76675-3. "The term used by American structuralists for **phonology**, indicating the central position of the phoneme in their analyses."
9. ^ Brentari, Diane; Fenlon, Jordan; Cormier, Kearsy (July 2018). "Sign Language Phonology". *Oxford Research Encyclopedia of Linguistics*. doi:10.1093/acrefore/9780199384655.013.117. ISBN 9780199384655. S2CID 60752232.
10. ^ Stokoe, William C. (1978) [1960]. *Sign Language Structure: An outline of the visual communication systems of the American deaf*. Department of Anthropology and Linguistics, University at Buffalo. Studies in linguistics, Occasional papers. Vol. 8 (2nd ed.). Silver Spring, MD: Linstok Press.
11. ^ "Definition of PHONOLOGY". *www.merriam-webster.com*. Retrieved 3 January 2022.
12. ^ *a b* Lass, Roger (1998). *Phonology: An Introduction to Basic Concepts*. Cambridge, UK; New York; Melbourne, Australia: Cambridge University Press. p. 1. ISBN 978-0-521-23728-4. Retrieved 8 January 2011Paperback ISBN 0-521-28183-0
13. ^ Carr, Philip (2003). *English Phonetics and Phonology: An Introduction*. Massachusetts, US; Oxford, UK; Victoria, Australia; Berlin, Germany: Blackwell Publishing. ISBN 978-0-631-19775-1. Retrieved 8 January 2011Paperback ISBN 0-631-19776-1
14. ^ *a b* Trubetzkoy N., *Grundzüge der Phonologie* (published 1939), translated by C. Baltaxe as *Principles of Phonology*, University of California Press, 1969
15. ^ Clark, John; Yallop, Colin; Fletcher, Janet (2007). *An Introduction to Phonetics and Phonology* (3rd ed.). Massachusetts, US; Oxford, UK; Victoria, Australia: Blackwell Publishing. ISBN 978-1-4051-3083-7. Retrieved 8 January 2011Alternative ISBN 1-4051-3083-0
16. ^ Bernards, Monique, "Ibn Jinnī", in: Encyclopaedia of Islam, THREE, Edited by: Kate Fleet, Gudrun Krämer, Denis Matringe, John Nawas, Everett Rowson. Consulted online on 27 May 2021 First published online: 2021 First print edition: 9789004435964, 20210701, 2021-4
17. ^ *a b* Anderson, Stephen R. (2021). *Phonology in the twentieth century* (Second, revised and expanded ed.). Berlin: Language Science Press. doi:10.5281/zenodo.5509618. ISBN 978-3-96110-327-0. ISSN 2629-172X. Retrieved 28 December 2021.
18. ^ Anon (probably Louis Havet). (1873) "Sur la nature des consonnes nasales". *Revue critique d'histoire et de littérature* 13, No. 23, p. 368.
19. ^ Roman Jakobson, *Selected Writings: Word and Language*, Volume 2, Walter de Gruyter, 1971, p. 396.
20. ^ E. F. K. Koerner, *Ferdinand de Saussure: Origin and Development of His Linguistic Thought in Western Studies of Language. A contribution to the history and theory of linguistics*, Braunschweig: Friedrich Vieweg & Sohn [Oxford & Elmsford, N.Y.: Pergamon Press], 1973.
21. ^ Hale, Mark; Reiss, Charles (2008). *The Phonological Enterprise*. Oxford, UK: Oxford University Press. ISBN 978-0-19-953397-8.
22. ^ Hale, Mark; Reiss, Charles (2000). "'Substance abuse' and 'dysfunctionalism': Current trends in phonology". *Linguistic Inquiry*. **31** (1): 157–169. JSTOR 4179099.
23. ^ Blevins, Juliette. 2004. *Evolutionary phonology: The emergence of sound patterns*. Cambridge University Press.
24. ^ Goldsmith 1995:1.

Bibliography [edit]

- Anderson, John M.; and Ewen, Colin J. (1987). *Principles of dependency phonology*. Cambridge: Cambridge University Press.
- Bloch, Bernard (1941). "Phonemic overlapping". *American Speech*. **16** (4): 278–284. doi:10.2307/486567. JSTOR 486567.
- Bloomfield, Leonard. (1933). *Language*. New York: H. Holt and Company. (Revised version of Bloomfield's 1914 *An introduction to the study of language*).
- Brentari, Diane (1998). *A prosodic model of sign language phonology*. Cambridge, MA: MIT Press.
- Chomsky, Noam. (1964). Current issues in linguistic theory. In J. A. Fodor and J. J. Katz (Eds.), *The structure of language: Readings in the philosophy language* (pp. 91–112). Englewood Cliffs, NJ: Prentice-Hall.
- Chomsky, Noam; and Halle, Morris. (1968). *The sound pattern of English*. New York: Harper & Row.
- Clements, George N. (1985). "The geometry of phonological features". *Phonology Yearbook*. **2**: 225–252. doi:10.1017/S0952675700000440. S2CID 62237665.
- Clements, George N.; and Samuel J. Keyser. (1983). *CV phonology: A generative theory of the syllable*. Linguistic inquiry monographs (No. 9). Cambridge, MA: MIT Press. ISBN 0-262-53047-3 (pbk); ISBN 0-262-03098-5 (hbk).
- de Lacy, Paul, ed. (2007). *The Cambridge Handbook of Phonology*. Cambridge University Press. ISBN 978-0-521-84879-4. Retrieved 8 January 2011.
- Donegan, Patricia. (1985). On the Natural Phonology of Vowels. New York: Garland. ISBN 0-8240-5424-5.
- Firth, J. R. (1948). "Sounds and prosodies". *Transactions of the Philological Society*. **47** (1): 127–152. doi:10.1111/j.1467-968X.1948.tb00556.x.
- Gilbers, Dicky; de Hoop, Helen (1998). "Conflicting constraints: An introduction to optimality theory". *Lingua*. **104** (1–2): 1–12. doi:10.1016/S0024-3841(97)00021-1.
- Goldsmith, John A. (1979). The aims of autosegmental phonology. In D. A. Dinnsen (Ed.), *Current approaches to phonological theory* (pp. 202–222). Bloomington: Indiana University Press.
- Goldsmith, John A. (1989). *Autosegmental and metrical phonology: A new synthesis*. Oxford: Basil Blackwell.
- Goldsmith, John A. (1995). "Phonological Theory". In John A. Goldsmith (ed.). *The Handbook of Phonological Theory*. Blackwell Handbooks in Linguistics. Blackwell Publishers. ISBN 978-1-4051-5768-1.
- Gussenhoven, Carlos & Jacobs, Haike. "Understanding Phonology", Hodder & Arnold, 1998. 2nd edition 2005.
- Hale, Mark; Reiss, Charles (2008). *The Phonological Enterprise*. Oxford, UK: Oxford University Press. ISBN 978-0-19-953397-8.
- Halle, Morris (1954). "The strategy of phonemics". *Word*. **10** (2–3): 197–209. doi:10.1080/00437956.1954.11659523.
- Halle, Morris. (1959). *The sound pattern of Russian*. The Hague: Mouton.
- Harris, Zellig. (1951). *Methods in structural linguistics*. Chicago: Chicago University Press.
- Hockett, Charles F. (1955). *A manual of phonology*. Indiana University publications in anthropology and linguistics, memoirs II. Baltimore: Waverley Press.
- Hooper, Joan B. (1976). *An introduction to natural generative phonology*. New York: Academic Press. ISBN 9780123547507.
- Jakobson, Roman (1949). "On the identification of phonemic entities". *Travaux du Cercle Linguistique de Copenhague*. **5**: 205–213. doi:10.1080/01050206.1949.10416304.
- Jakobson, Roman; Fant, Gunnar; and Halle, Morris. (1952). *Preliminaries to speech analysis: The distinctive features and their correlates*. Cambridge, MA: MIT Press.
- Kaisse, Ellen M.; and Shaw, Patricia A. (1985). On the theory of lexical phonology. In E. Colin and J. Anderson (Eds.), *Phonology Yearbook 2* (pp. 1–30).
- Kenstowicz, Michael. *Phonology in generative grammar*. Oxford: Basil Blackwell.
- Ladefoged, Peter. (1982). *A course in phonetics* (2nd ed.). London: Harcourt Brace Jovanovich.
- Martinet, André (1949). *Phonology as functional phonetics*. Oxford: Blackwell.
- Martinet, André (1955). *Économie des changements phonétiques: Traité de phonologie diachronique*. Berne: A. Francke S.A.
- Napoli, Donna Jo (1996). *Linguistics: An Introduction*. New York: Oxford University Press.
- Pike, Kenneth Lee (1947). *Phonemics: A technique for reducing languages to writing*. Ann Arbor: University of Michigan Press.
- Sandler, Wendy and Lillo-Martin, Diane. 2006. *Sign language and linguistic universals*. Cambridge: Cambridge University Press
- Sapir, Edward (1925). "Sound patterns in language". *Language*. **1** (2): 37–51. doi:10.2307/409004. JSTOR 409004.
- Sapir, Edward (1933). "La réalité psychologique des phonémes". *Journal de Psychologie Normale et Pathologique*. **30**: 247–265.
- de Saussure, Ferdinand. (1916). *Cours de linguistique générale*. Paris: Payot.
- Stampe, David. (1979). *A dissertation on natural phonology*. New York: Garland.
- Swadesh, Morris (1934). "The phonemic principle". *Language*. **10** (2): 117–129. doi:10.2307/409603. JSTOR 409603.
- Trager, George L.; Bloch, Bernard (1941). "The syllabic phonemes of English". *Language*. **17** (3): 223–246. doi:10.2307/409203. JSTOR 409203.
- Trask, Robert Lawrence, ed. (1996). *A Dictionary of Phonetics and Phonology*. Routledge. ISBN 0-415-11260-5.
- Trubetzkoy, Nikolai. (1939). *Grundzüge der Phonologie*. Travaux du Cercle Linguistique de Prague 7.
- Twaddell, William F. (1935). On defining the phoneme. Language monograph no. 16. *Language*.

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